Name: _____ ( ) Class: 26 / _____

ANDERSON SERANGOON JUNIOR COLLEGE

2026 JC2 Preliminary Examination

PHYSICS Higher 2 9478/03

Paper 3 Longer Structured Questions **Thursday 27 August 2026**

2 hours

You must answer on the question paper.

No additional materials are required.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, class index number and class in the spaces at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in the brackets [ ].

For Examiner's Use	
Paper 3 (75 marks)	
1	
2	
3	
4	
5	
6	
7	
8	
9	
Deductions	
Total	

This document has **28** pages.

Data

speed of light in free space	$c = 3.00 \times 10^8$ m s^{-1}
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}
	$(\frac{1}{4\pi\varepsilon_o} = 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on/by a gas	$W = p\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = -\frac{Gm}{r}$
temperature	T / K = T/°C + 273.15
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$

displacement of particle in s.h.m.	$x = x_o \sin \omega t$
velocity of particle in s.h.m.	$v = v_o \cos \omega t = \pm\omega\sqrt{x_o^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + ...$
capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + ...$
capacitors in parallel	$C = C_1 + C_2 + \ldots$
energy in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	$Q = Q_o\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	$Q = Q_o e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_o r}$
alternating current / voltage	$x = x_o \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_o I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_o N I}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 n I$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2}n^2$
radioactive decay	$x = x_o e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

Section A

Answer **all** the questions in this section in the spaces provided.

1 **(a)** Explain why the unit of pressure is a derived unit.

……..

……..

…….....[2]

(b) A physical equation is of the form

$$Q = \frac{kr^4 (p_1 - p_2)}{L}$$

where k is a dimensionless constant,

r = (2.07 ± 0.04) mm,

p_1 = (127 ± 2) kPa,

p_2 = (105 ± 1) kPa,

L = (200 ± 8) mm.

Calculate the fractional uncertainty in Q.

fractional uncertainty in Q = …… [3]

[Total: 5]

Turn over

2 **(a)** Define the moment of a force about a point.

……

…… [1]

(b) A tree of mass 270 kg grows out of sloping ground and is supported by a post, as shown in Figure 2.1.

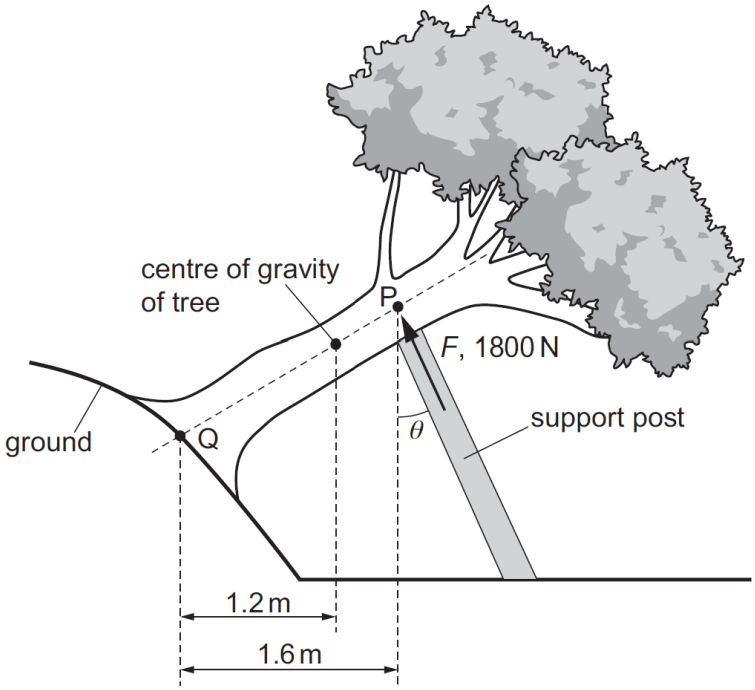


Figure 2.1 (not to scale)

The ground applies a total force *R* on the tree at point Q. The centre of gravity of the tree is a horizontal distance of 1.2 m from Q.

The post applies a force *F* of 1800 N perpendicular to the line PQ. The line of action of *F* passes through point P at an angle *θ* to the vertical. P is a horizontal distance of 1.6 m from Q.

The tree is in equilibrium and all forces act on the tree in the same plane.

(i) Calculate angle *θ*.

θ = ……° [3]

(ii) State and explain whether the total force *R* at point Q is horizontal.

……

……

…… [2]

[Total: 6]

3 A bungee jumper of mass 64 kg secures one end of an elastic rope to a bridge. The other end is attached to the bungee jumper. The jumper falls from rest from the bridge and descends into the valley below, as shown in Figure 3.1.

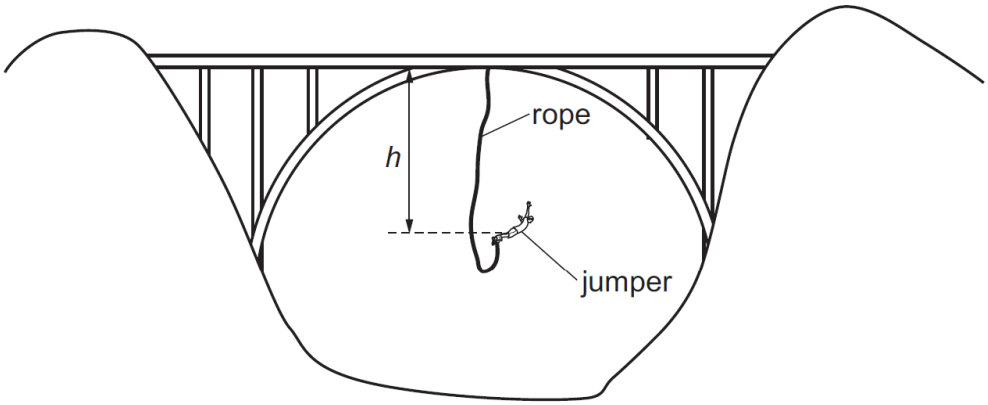


Figure 3.1 (not to scale)

Figure 3.2 shows the variation of the tension T in the rope with the vertical distance h of the jumper below the level of the bridge.

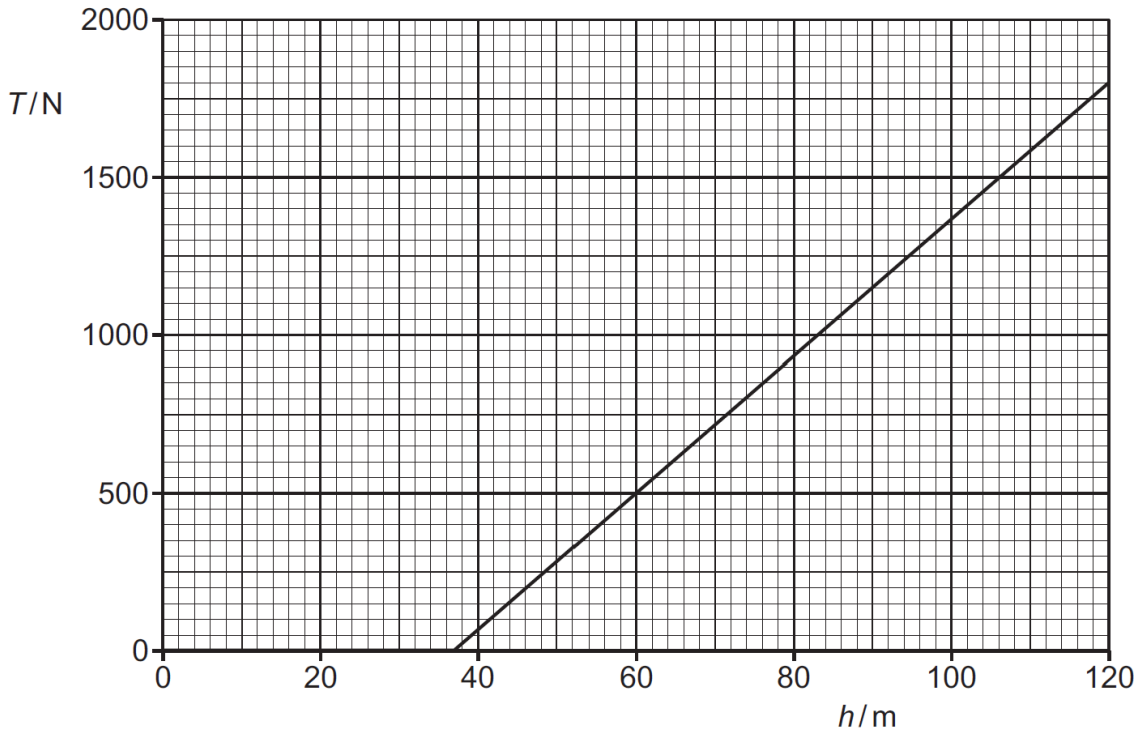


Figure 3.2

(a) The position of the bungee jumper is at a distance of 120 m below the bridge.

(i) Show that the loss of gravitational potential energy since leaving the bridge is 75 kJ.

[1]

(ii) Calculate the elastic potential energy in the rope.

elastic potential energy = …….J [2]

(b) Explain what can be deduced from the information in **3(a)** about the speed of the bungee jumper when at a distance of 120 m below the bridge.

…….....

…….....

……..... [2]

[Total: 5]

4 **(a)** A small ball rests at point P on a curved track of radius *r*, as shown in Figure 4.1.

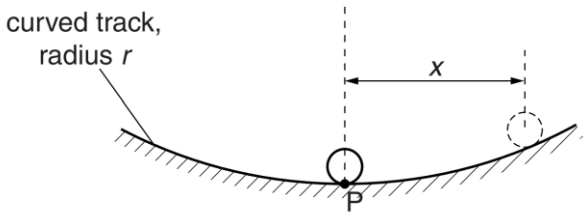


Figure 4.1

The ball is moved a small distance to one side and is then released. The horizontal displacement *x* of the ball is related to its acceleration *a* towards P by the expression

$$a = -\frac{gx}{r}$$

where *g* is the acceleration of free fall.

(i) Show that the ball undergoes simple harmonic motion.

……

……

……

…… [2]

(ii) The radius *r* of curvature of the track is 28 cm.

Determine the time interval τ between the ball passing point P and then returning to point P.

τ = ... s [3]

(b) The variation with time t of the displacement x of the ball in **4(a)** is shown in Figure 4.2.

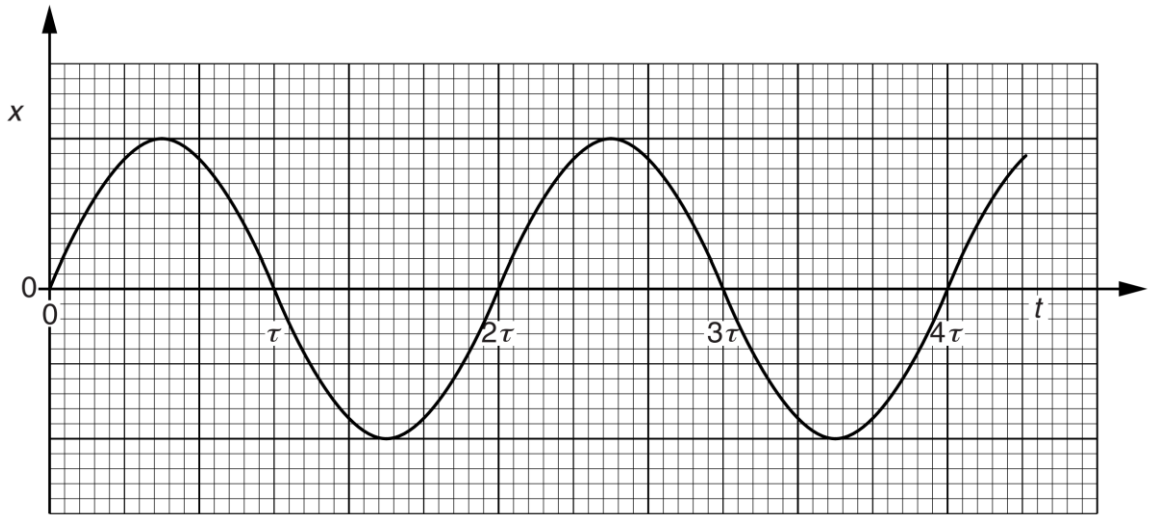


Figure 4.2

Some moisture now forms on the track, causing the ball to come to rest after approximately 15 oscillations.

On the axes of Figure 4.2, sketch the variation with time t of the displacement x of the ball for the first two periods after the moisture has formed. Assume the moisture forms at time $t = 0$.

[3]

[Total: 8]

5 **(a)** A magnet produces a uniform magnetic field of flux density *B* in the space between its poles.

A rigid copper wire carrying a current is balanced on a pivot. Part PQLM of the wire is between the poles of the magnet, as illustrated in Figure 5.1.

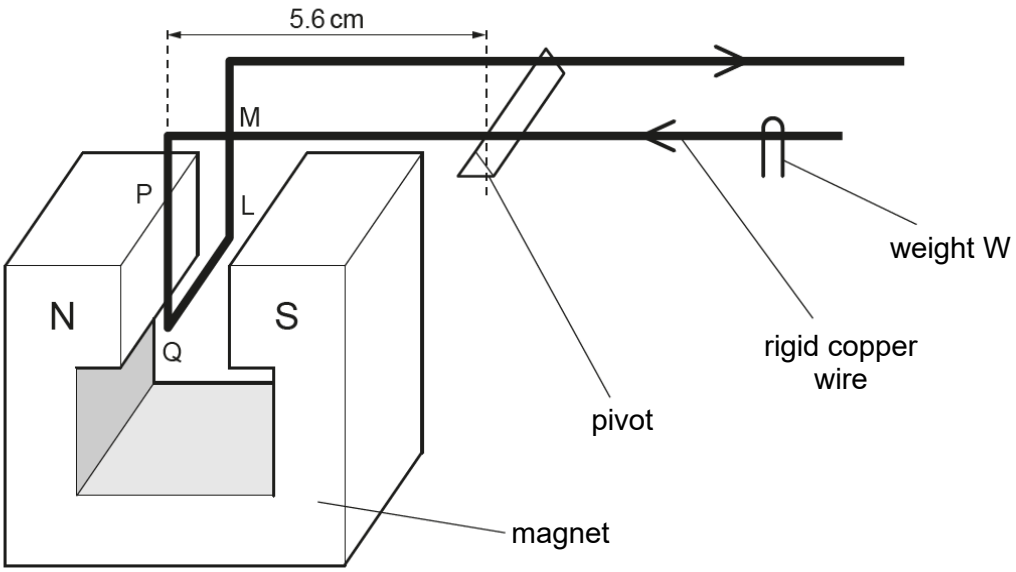


Figure 5.1 (not to scale)

The wire is balanced horizontally by a small weight W whose mass is 1.3×10^{-4} kg.

Section QL of the wire has length 0.85 cm.

The perpendicular distance of QL from the pivot is 5.6 cm.

When the current in the wire is changed by 1.2 A, W is moved a distance of 2.6 cm along the wire in order to restore equilibrium.

(i) Show that the change in moment of W about the pivot is 3.3×10^{-5} N m.

[1]

(ii) Use the information in **5a(i)** to determine the magnetic flux density *B* between the poles of the magnet.

B = ……T [2]

(b) In a different experiment, electrons are accelerated through a potential difference and then enter a region of uniform magnetic field of flux density 1.5 mT.

Figure 5.2 illustrated such an electron travelling at right angles to the uniform magnetic field.

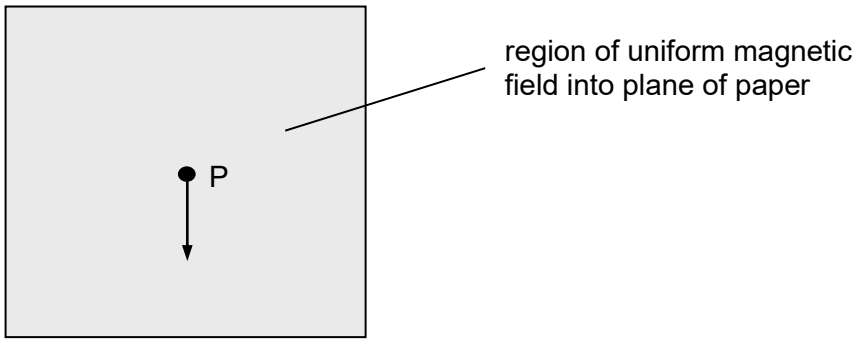


Figure 5.2

The magnetic field is directed into the plane of the paper.

When the electron is at P, its velocity is 2.9×10^7 m s^{-1} in the direction shown. This is normal to the magnetic field.

(i) On Figure 5.2, sketch the path of the electron, assuming that it does not leave the region of the magnetic field.

[2]

(ii) Calculate the radius of the path of the electron. Explain your working.

radius = ……..m [2]

(iii) A uniform electric field is now produced in the same region and in the same direction as the magnetic field. Suggest the shape of the resultant path of the electron. You may draw a sketch to illustrate the path, if you wish.

……

……

…… [2]

[Total: 9]

6 A pair of concentric coils is shown in Figure 6.1.

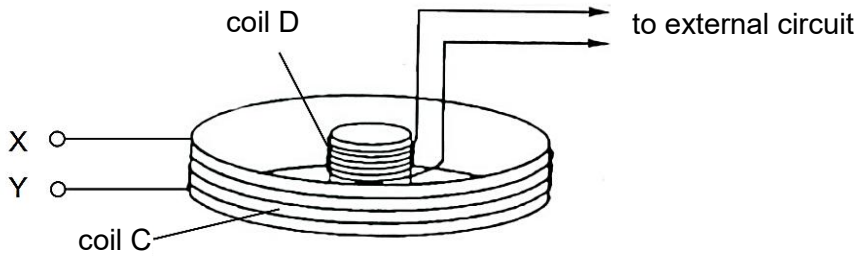


Figure 6.1

The outer coil C is connected to a variable power supply by the terminals XY.

The variation with time t of the current I in coil C is shown in Figure 6.2.

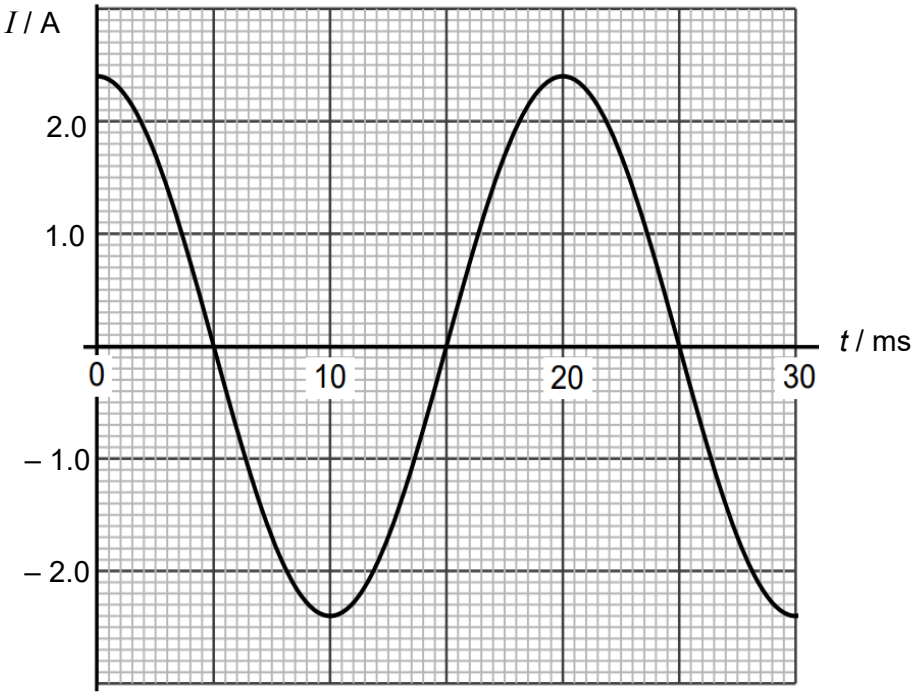


Figure 6.2

(a) Use Faraday's law to explain why an electromotive force (e.m.f.) is induced in coil D.

……

……

……

……

……

…… [2]

(b) State an instant when the induced e.m.f. in coil D is a maximum value.

t = …….ms [1]

(c) Explain your answer for **6(b)**.

…….....

…….....

……..…… [1]

(d) The outer coil C produces a uniform magnetic field of flux density B, in tesla, in the region of coil D given by the expression

$$B = 1.30 \times 10^{-3} I$$

The coil D has a circular area of 1.05×10^{-3} m^2 and contains 50 turns of wire.

From t = 10 ms to t = 15 ms,

(i) calculate the average induced e.m.f in coil D,

e.m.f. = …….V [3]

(ii) use Lenz's Law to explain why the induced current in coil D is in the same direction as the current in coil C.

…….....

…….....

…….....

…….....

…….....[2]

[Total: 9]

7 **(a)** A sample of a radioactive isotope contains N nuclei of the isotope at time T.
At time $(T + \Delta T)$, the sample contains $(N - \Delta N)$ nuclei of the isotope. The time interval ΔT is short.

Use the symbols N, ΔN, T and ΔT to give expressions for:

(i) the average activity of the sample during the time ΔT

..... [1]

(ii) the probability of decay of a nucleus in the time ΔT

..... [1]

(iii) the decay constant λ of the isotope.

..... [1]

(b) The isotope polonium-208 (${}^{208}_{84}\mathrm{Po}$) is radioactive and decays to form lead-204 (${}^{204}_{82}\mathrm{Pb}$).
The nuclear equation for this decay is

$$^{208}_{84}\mathrm{Po} \rightarrow {}^{204}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$$

Data for the nuclei in the reaction are given in Table 7.1.

Table 7.1

nucleus	mass / u
${}^{4}_{2}\mathrm{He}$	4.002603
${}^{204}_{82}\mathrm{Pb}$	203.973043
${}^{208}_{84}\mathrm{Po}$	207.981245

(i) Determine, to three decimal places, the energy released, in MeV, associated with the decay of polonium-208.

energy = ... MeV [3]

(ii) The polonium-208 nucleus is initially stationary. The initial kinetic energy of the $^{4}_{2}He$ nucleus (α-particle) is found to be less than the energy calculated in **7b(i)**.

Suggest **two** possible reasons for this difference.

1. ……

……..

2. ……

……..

[2]

(c) Lanthanium-140 ($^{140}_{57}La$) emits a β-particle to become a stable nuclide, cerium (Ce). The decay is represented by the equation

$$^{140}_{57}La \rightarrow \beta + Ce$$

(i) State the nuclear notation for cerium.

$$^{\cdots\cdots}_{\cdots\cdots}Ce$$

[1]

(ii) Lanthanum-140 has a half-life of $T_{1/2}$.

A pure sample of lanthanum-140 contains N_0 nuclei.

The variation with time t of the number N of lanthanum-140 nuclei is shown in Figure 7.1.

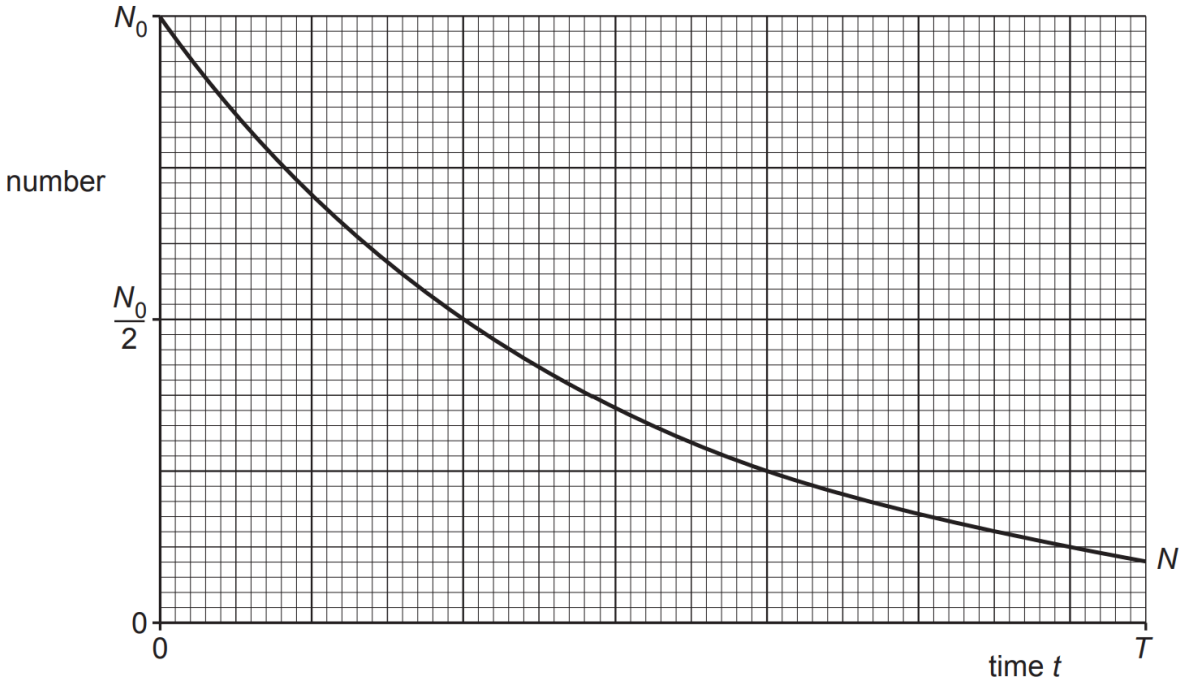


Figure 7.1

On Figure 7.1:

- label, on the time axis, the time $t = 1.0T_{1/2}$ and the time $t = 2.0T_{1/2}$.
- sketch the variation with time t of the number of nuclei of cerium for $t = 0$ to $t = T$.

[2]

(d) In 2024, scientists in Montana discovered a fossil of the dinosaur, *Lokiceratops*.

To find the age of the fossil, scientists measured a radioactive isotope in a mineral sample taken from the volcanic ash layer surrounding the fossil.

The half-life of the isotope is 7.80×10^7 years.

A 12 g sample of new mineral sample from fresh volcanic rock has an activity of 36 Bq.

A 4 g sample of ancient mineral from the fossil's ash layer has an activity of 5.0 Bq.

Calculate the age of the fossil.

age = ... years [2]

[Total: 13]

Turn over

Section B

Answer **one** question from this section in the spaces provided.

8 A hot air balloon begins to ascend vertically from level ground at time t = 0 s.

Figure 8.1 shows the variation with t of the velocity v of the hot air balloon during its ascent from t = 0 to t = 25 s.

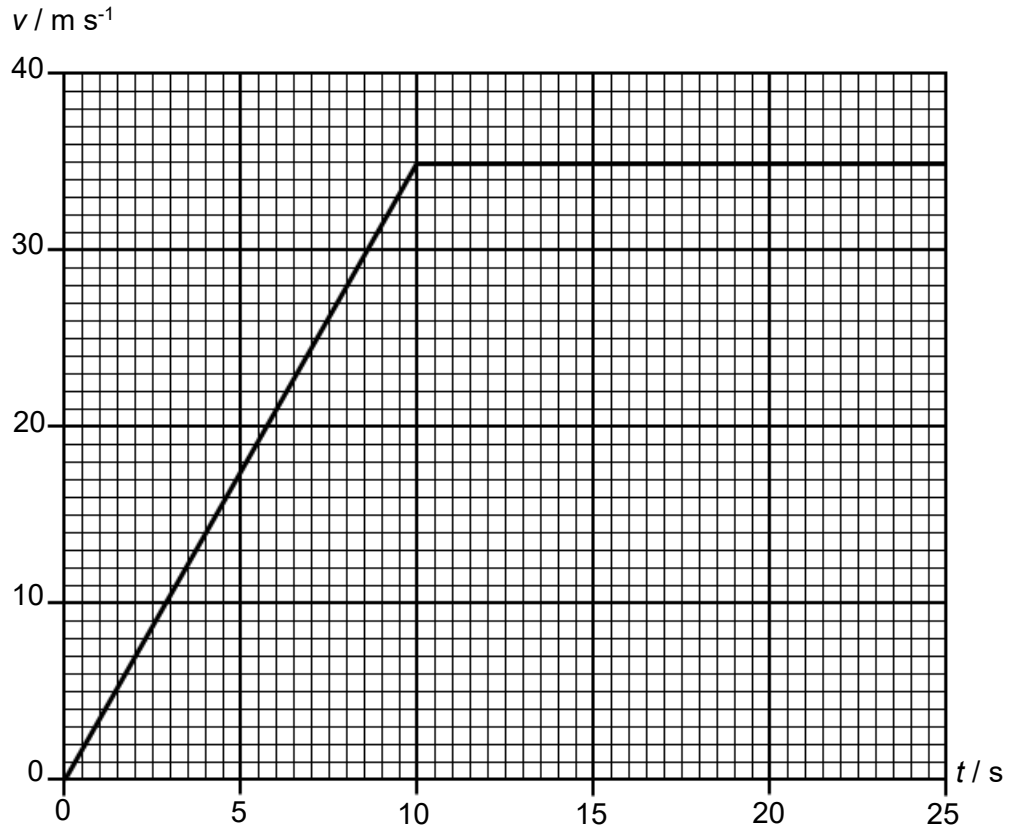


Figure 8.1

(a) Using Figure 8.1, calculate the acceleration of the hot air balloon from t = 0 to t = 10 s.

acceleration = ... m s^{-2} [1]

(b) At t = 3.0 s, a boy in the basket of the balloon realises that he has left his phone on the ground.

His friend, who is standing on the ground, throws the phone vertically upwards at this instant with a speed of 30 m s^{-1}.

The speed is insufficient for the phone to reach the boy.

(i) On Figure 8.1, sketch the variation with t of the velocity of the phone until the velocity is zero. [1]

(ii) Hence or otherwise, estimate the minimum separation between the boy and the phone.

separation = …… m [3]

(c) At t = 15.0 s, the boy drops a ball vertically downwards from the basket of the balloon.

Air resistance is negligible.

(i) Calculate the time taken for the ball to reach the ground after it is dropped.

time = …… s [2]

(ii) On Figure 8.2, sketch the variation with time t of the vertical displacement s of the ball from the instant it is released until it hits the ground.

Label this graph A. Include known values on the graph. [2]



Figure 8.2

(iii) At the same instant t = 15.0 s, the boy also releases a plastic ball for which air resistance is not negligible.

On Figure 8.2, sketch the variation with time t of the vertical displacement s of the plastic ball.

Label this graph B.
Assume that the plastic ball reaches terminal speed before hitting the ground.

[3]

(d) Another ball of mass 500 g, thrown by the boy, hits the ground at a speed of 11.5 m s^{-1} at an angle of 50° to the horizontal, as shown in Figure 8.3.

The ball rebounds with speed v at an angle of 35° to the horizontal.

Assume that the ground is frictionless.

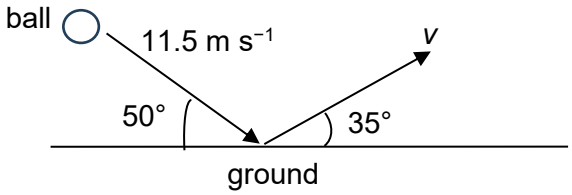


Figure 8.3

(i) Show that the rebound speed v of the ball is 9.0 m s^{-1}.

[2]

(ii) Calculate the magnitude of the change in velocity of the ball.

change in velocity = ... m s^{-1} [2]

(iii) The ball is in contact with the ground for a duration of 0.20 s.

Determine the magnitude of the average force exerted on the ball by the ground.

force = ... N [2]

(iv) State and explain how the rebound speed would change if friction between the ball and the ground were not negligible.

……

……

…… [2]

[Total: 20]

9 **(a)** A resistance wire of uniform cross-sectional area 3.3×10^{-7} m^2 and length 2.0 m is made of metal of resistivity 5.0×10^{-7} Ω m.

(i) Determine the resistance of the wire.

resistance = ……..Ω [1]

(ii) The ends of the resistance wire are connected to the terminals S and T in the circuit shown in Figure 9.1.

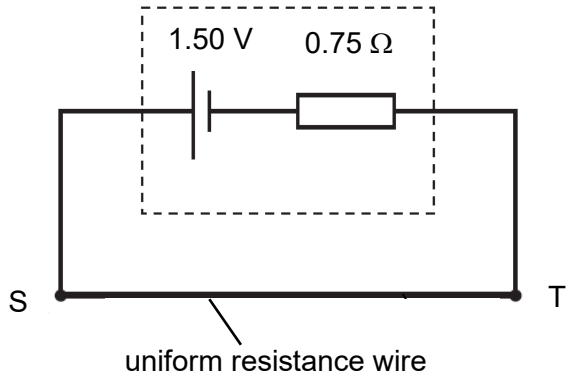


Figure 9.1

The cell has an e.m.f. of 1.50 V and internal resistance 0.75 Ω.

The number density of charge carriers in the wire is 8.5×10^{28} m^{-3}.

Calculate the average drift speed of the charge carriers.

speed = ……..m s^{-1} [2]

(iii) The wire in **9(a)(ii)** is now replaced with another wire of the same material and diameter but twice the length. The temperature of both wires is the same.

State and explain the change, if any, to the drift velocity of the charge carriers in the second wire.

……

……

……

……

……[3]

(b) The variation with potential difference V of current I for a light emitting diode (LED) is shown in Figure 9.2.

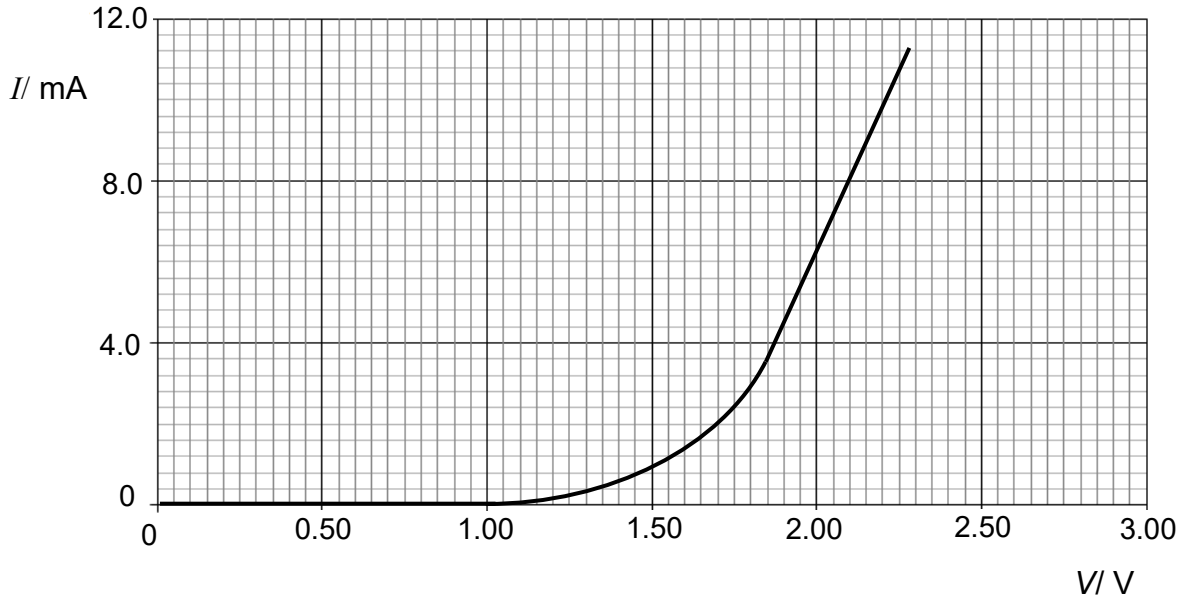


Figure 9.2

(i) Use Figure 9.2 to determine the resistance of the LED at 2.25 V.

resistance = ……Ω [2]

(ii) On Figure 9.2, draw the $I - V$ characteristics of a 160 Ω ohmic resistor. Label the line R.

[1]

(iii) Two of these LEDs are connected to a 3.0 V battery with negligible internal resistance and an ohmic 160 Ω resistor as shown in Figure 9.3.

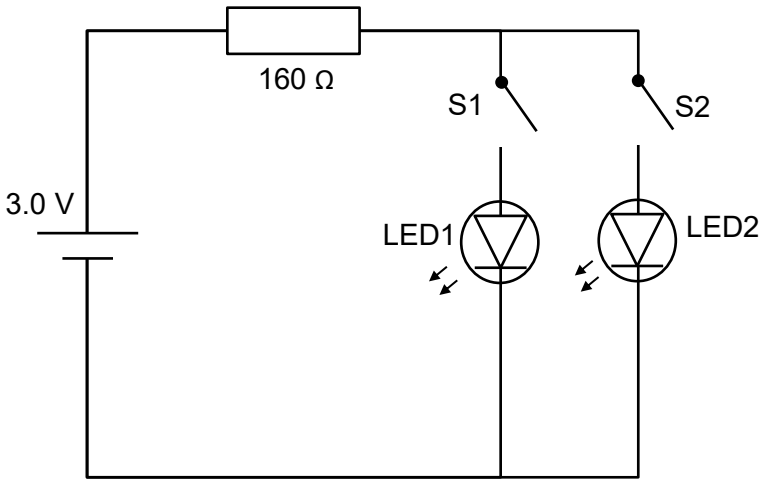


Figure 9.3

1. Determine the current in LED1 when switch S1 is closed and switch S2 remains open. Show your working clearly.

current = …….A [2]

2. Explain why LED1 becomes dimmer when S2 is also closed.

…….....

…….....

…….....

…….....

……..……..[3]

(c) A sinusoidal alternating voltage supply, with a peak value of 150 V, is applied across the primary coil of a transformer. There are 1200 turns in the primary coil. The variation with time *t* of the e.m.f *E* induced across the secondary coil is shown in Figure 9.4.

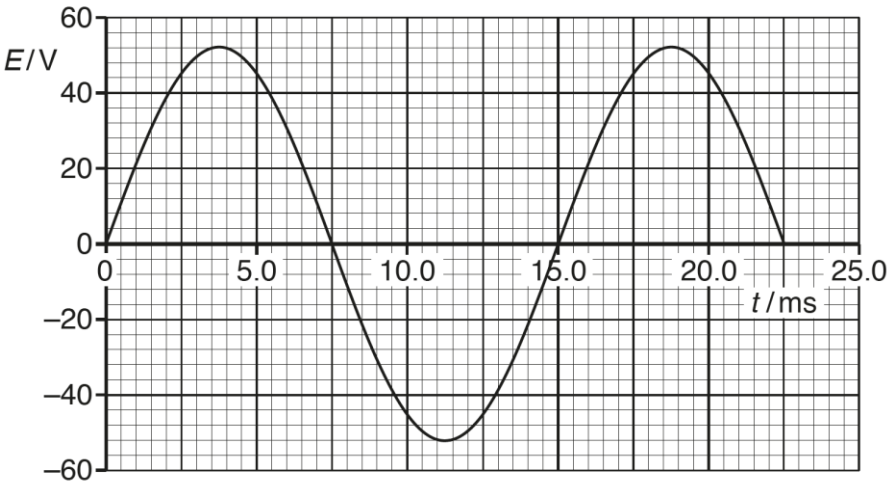


Figure 9.4

(i) Use data from Figure 9.4 to calculate the number of turns in the secondary coil.

number of turns = …….…….. [2]

(ii) A resistor is connected between the output terminals of the secondary coil. The mean power dissipated in the resistor is 1.2 W. It may be assumed that the varying voltage across the resistor is equal to the varying e.m.f. *E* shown in Figure 9.4.

1. Calculate the resistance of the resistor.

resistance = ……Ω [2]

2. On Figure 9.5, sketch the variation with time t of the power P dissipated in the resistor for t = 0 to t = 22.5 ms.

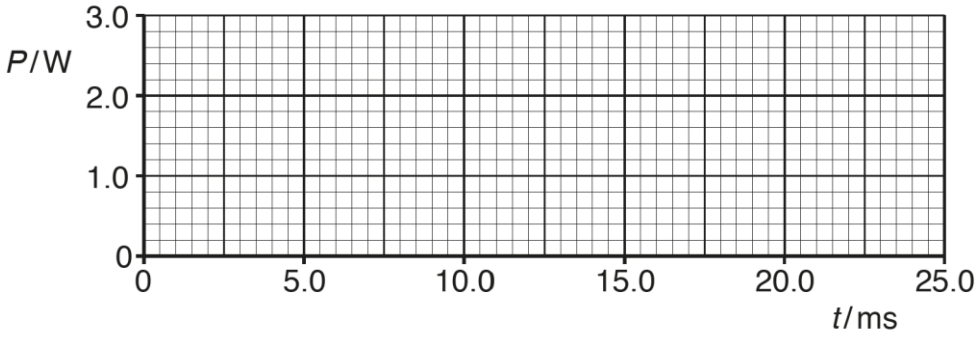


Figure 9.5

[2]

[Total: 20]